A Report on

"Multi-LLM Deliberation Framework for Research Paper Analysis and Question Answering"

MASTER OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted by

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GitHub Repository Link : https://github.com/sutarshoeb/llm-council

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CERTIFICATE

This is to certify that the Project work titled "**Multi-LLM Deliberation Framework for Research Paper Analysis and Question Answering**" by **Shoeb Sutar** under the course Large Language Models under **Master of Technology** in **Artificial Intelligence & Machine Learning**, Symbiosis International (Deemed University), Pune during the academic year 2025 - 2026.

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DECLARATION

I hereby declare that the project titled **“Multi-LLM Deliberation Framework for Research Paper Analysis and Question Answering”**, submitted to the Symbiosis Institute of Technology, Pune, in partial fulfillment of the requirements for the degree of **Master of Technology in Artificial Intelligence & Machine Learning**, is the outcome of my original work. I also affirm that this project report, or any part of it, has not been submitted previously to any other University or Institute for the award of any degree or diploma.

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Date : 24 / 04 / 2026

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ABSTRACT

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding, summarization, and question answering. However, single models suffer from hallucinations, model-specific bias, and absence of self-evaluation, these limitations are particularly very critical in academic and research contexts. This study involves a LLM Council which is a multi LLM deliberation framework applied specifically to the research paper analysis and academic question answering. The system sends the user queries to a council made of four OpenAI models at the same time, anonymizes their responses to eliminate self-preference bias, and uses a meta-model as an impartial judge that evaluates each response across four academic criteria such as Correctness, Clarity, Completeness, and Academic Depth before synthesizing a final answer. The framework is based on two well-established principles : ensemble learning and academic peer review. The system is implemented using the OpenAI Responses API with native support for PDF. The system supports three input modes such as text, URL, and file upload, making it suitable for a wide range of research assistance tasks. Results show that multi-LLM deliberation produces responses that are more reliable and comprehensive than a single model, offering a promising direction for building trustworthy AI-powered research tools.

Keywords - Large Language Models, Multi-Model Ensemble, LLM-as-a-Judge, Research Assistance, Deliberation Framework, Prompt Engineering

1. INTRODUCTION

The development of Large Language Models (LLMs) has changed the way humans are able to engage with information. The existing models have demonstrated remarkable capabilities in the field of natural language understanding, summarization, and question answering. However, a fundamental limitation exists, when a user asks a query to a single LLM, the response given by model is limited by the data on which model is trained, the bias that may be present in data, and reasoning patterns. This limitation of model becomes critical in academic and research contexts, where precision, completeness, and factual accuracy and correctness of solution are nonnegotiable. A single model may produce an incomplete or subtly incorrect summary with high confidence, with no existing mechanism to verify or cross-check the output given by the model.

The study presents the LLM Council, which is a multi-model deliberation framework inspired by Andrej Karpathy's LLM Council project [2]. This framework is implemented independently from scratch and applied specifically to the domain of research paper analysis and academic question answering. The system sends a user's research query to multiple specialized LLMs simultaneously, collects their independent responses, anonymizes these responses, and passes them to a meta-model that acts as an impartial judge which will be evaluating, ranking, and synthesizing every response based on pre-defined rubric and generate a final consolidated response to query.

The key idea behind this study is simple : no single LLM is universally best, but a structured council of diverse models, each of which is contributing its strengths, can finally produce a response that is much more accurate, complete, and reliable than any individual models answer

2. RELATED WORK

Single-model LLM pipelines suffer from three core problems in context of research assistance :

1. LLMs are known to generate text which is plausible-sounding but factually incorrect, particularly when we are dealing with very specific research topics, recent research papers, or highly technical content. There is no built-in verification layer in standard single-model systems.
2. Every LLM is a result of its training data, Reinforcement Learning from Human Feedback (RLHF) alignment, and architectural choices. A model which is trained on web data may overly simplify academic related content. A reasoning focused model may over explain simple concepts. These biases are added to the model or sometimes may amplify in the model. These biases are invisible to the end user especially in case of a single model.
3. The standard LLM pipelines generate a response and stop. There is no mechanism for the model to compare its answer against alternative response or interpretation, assess its own weaknesses, or synthesize results from multiple perspectives, all of which are standard practices in human academic peer review.

In summary, the problem is, how can we design an LLM based system that mimics the collaborative, peer-reviewed nature of academic discourse to produce more reliable research assistance ?

3. PROPOSED SOLUTION

The LLM Council for Research Assistance addresses the limitations of a single model pipeline by introducing a three stage multi model deliberation architecture. Instead of relying on a single model to answer a research query, the system creates a council of four domain-specialized LLMs, each of which independently analyzes the query, followed by a meta-model that will evaluate all the responses and synthesize their responses.

The solution draws inspiration from two well-established concepts. Ensemble Learning — a classical machine learning technique where multiple models vote or combine their outputs to produce a result more accurate and robust than any individual model. The LLM Council applies this principle at the level of generative language models rather than traditional machine learning models. Academic Peer Review - the normal procedure in academic publishing whereby before a submission is accepted, it is reviewed by a separate reviewer to check its accuracy, clarity and completeness. The meta-model of this system does a comparable task of being an impartial judge and the evaluation of anonymized responses of models. The two principles taken together form a system which is both more accurate and robust than a single LLM and more principled than the mere aggregation of the products of the models.

4. SYSTEM OVERVIEW

The system is implemented in Python, and uses the OpenAI Response API as a single unified gateway to access multiple models. The overall pipeline can be summarized as follows :

Fig.1. LLM Council Pipeline

The system is structured around five core functions, each of which is responsible for one logical stage of the entire pipeline :

Table 1 : Core functions of the LLM Council system

Function Name	Responsibility
`call_llm()`	Single API call; returns model, content, latency, status
`query_llm()`	Dispatches query to all council models in parallel
`file_upload()`	Uploads local PDF to OpenAI Files API, returns `file_id`
`blind_responses()`	Anonymizes responses with neutral labels
`final_judge_response()`	Sends blind responses to meta-model for evaluation
`run_llm_council()`	Master orchestration; runs full pipeline end-to-end

5. SYSTEM ARCHITECTURE

5.1 Configuration Layer

The topmost layer which defines all the constants that control the behavior of the entire pipeline. This includes the council model registry (`MODELS`), the meta-model identifier `META_MODEL`, the system prompt `SYSTEM_PROMPT` and the temperature parameter (`TEMPERATURE`). The council of LLMs consists of four research specialized models, each of which is selected for a distinct academic strength. The meta-model (`gpt-5.4`) acts as the Chairman, it does not answer the question itself but evaluates and synthesizes the answers provided by the council members.

Table 2 : Council models and their research specializations

Model	Type	Research Strength
`gpt-5.4-mini`	Language Model	Fast, efficient general QA
`o3`	Reasoning Model	Deep multi-step academic reasoning
`gpt-5.1`	Language Model	Balanced research analysis
`gpt-5.2`	Language Model	Strong academic writing & synthesis

5.2 Inference Layer

This layer of system is responsible for sending the user query to all council models simultaneously and collecting their responses. The key design decision here is parallel inference using Python threading. Rather than calling each model sequentially, which would result in cumulative latency, the system creates one thread per model using `threading.Thread`. All four models are sent the query simultaneously, and the main thread waits for all to complete using `thread.join()`. This reduces total response time from the sum of individual latencies to approximately the latency of the slowest model. All API calls are made through the OpenAI Responses API (`client.responses.create()`), which natively supports file inputs via `file_id` or `file_url`, enabling direct PDF ingestion without any need of an external extraction library. Failure in any calls are also handled gracefully. If a model is unavailable or fails to generate response, it is excluded from the council output without crashing the pipeline.

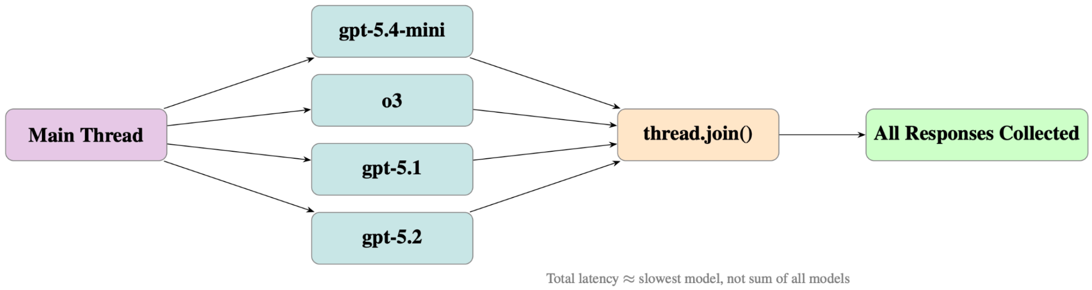

Fig.2. Parallel threading in `query_llms()`

5.3 Anonymization Layer

Before the responses are passed to the meta-model, they go through a critical intermediate step, blind anonymization. The `blind_responses()` function strips all model identity information and relabelles each successful response with a neutral alphabetic identifier, Response A, Response B, Response C, Response D.

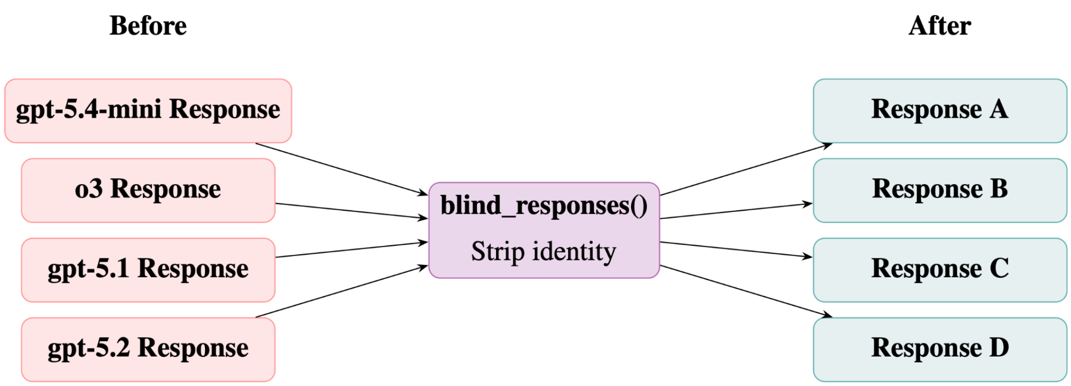

Fig.3. Anonymization of model responses using `blind_responses()`

This design decision directly addresses a known failure mode in LLM-as-a-judge systems called self-preference bias, which is the tendency of a model to rate its own output higher and better than others when it can identify which model generated which response. By anonymizing responses, it ensures that the meta-model evaluates only on the basis of quality of content a model has generated.

5.4 Deliberation Layer

The final layer is where the synthesis happens. The meta-model receives three inputs, the original user query, the anonymized responses, and a structured judge prompt. The judge prompt clearly instructs the meta-model to evaluate each response across four academic criteria.

1. **Correctness :** Factual accuracy of the content
2. **Clarity :** Structure and readability of the response
3. **Completeness :** Coverage of methodology, findings, and limitations
4. **Academic Depth :** Ability to go beyond surface-level explanation

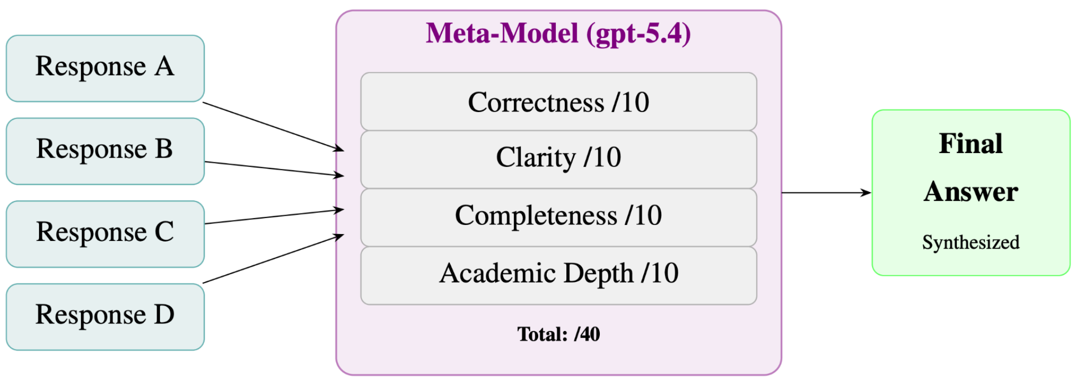

Fig.4. Judge scoring framework in the deliberation layer

6. MAPPING WITH LLM CONCEPTS

6.1 Autoregressive Language Modeling & Temperature Sampling

At the very core of every council member is a LLM which generates responses token by token. At each step, the model computes a probability distribution over its learned vocabulary and selects the next token. The `temperature` parameter is used to control the sharpness of this distribution. Setting `temperature = 0.7` across all council members ensures responses are neither too rigid nor too random which is essential for academic content which requires both accuracy and fluency.

6.2 Attention Mechanism & Long-Context Handling

Each model is an Decoder-only autoregressive model which is built on the Transformer architecture introduced by [1], where the self-attention mechanism allows the model to relate to every token in the input to every other token. For research queries, which may include long paper abstracts or multi-part questions, this is very important. The attention score between query token q and key token k is computed as :

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Models such as gpt-5.2 and o3 can take advantage of longer context windows, enabling them to read a whole research paper at once, which is the key feature in the use case of research assistance [4].

6.3 Ensemble Learning Applied to LLMs

Classical ensemble learning combines multiple weaker models to produce a stronger collective output [5]. The LLM Council applies this same principle at the generative model level. Rather than training multiple models, it assembles OpenAI models of two distinct architectural families, reasoning models (o3) trained with reinforcement learning for multi-step problem solving, and language models like (GPT-5.x) family trained for fluent generation of text which ensures that architectural diversity within a single provider ecosystem

6.4 LLM-as-a-Judge & Prompt Engineering

The role of meta-model in this system is a direct implementation of the LLM-as-a-Judge paradigm, using a language model to evaluate the outputs of other language models. The quality of this evaluation is entirely determined by the judge prompt, making prompt engineering a critical NLP component of this system. The judge prompt is structured to go through four specific evaluation dimensions, forcing it to evaluate in a numeric scoring format, and request a final synthesized answer, all of this in a single inference call. This is an application of instruction following, where the format of output of the model is controlled entirely through natural language directives rather than code [3].

6.5 Anonymization & Bias Mitigation

The `blind_responses()` function implements a form of debiasing at the inference level. Research in LLM evaluation has shown that models exhibit measurable self-preference bias where model rates their own outputs higher when they can identify authorship. By replacing model names with neutral labels (Response A, B, C, D), the system eliminates this bias without any additional fine-tuning or post-processing, relying purely on information given by a model [6].

7. EVALUATION METRICS

Evaluation in the LLM Council operates at two levels, evaluating the quality of individual model responses, and evaluating the reliability of the meta-model itself.

7.1 Response Quality Metrics

Each anonymized response is evaluated by the meta-model using four criteria, each rated out of 10, up to a total of 40 points based on the sum of the scores the model will be ranked on each query and can be compared in a formatted evaluation table.

Table 3 : Response Quality Metrics

Metric	Description
Correctness	Factual accuracy of the response content
Clarity	Structure, readability, and coherence
Completeness	Coverage of methodology, findings, limitations
Academic Depth	Ability to go beyond surface-level explanation

7.2 Position Bias Test

To evaluate reliability of meta-model, the same query can be run twice with the blind responses shuffled in a different order. If the meta-model consistently selects the same response as best regardless of its position in the list, it demonstrates position invariant judgment which is a desirable property in any evaluation system. Inconsistency across runs would indicate position bias, a known failure mode in LLM-as-a-Judge systems.

7.3 Human Agreement Rate

A lightweight human evaluation can be conducted by having the evaluator independently read all model responses and select the one that is best, then comparing that choice with the selection of meta model. Agreement across 5 test queries produces a human agreement rate which is a simple but meaningful measure of the meta-model's alignment with human academic judgment.

8. APPLICATION AND FUTURE SCOPE

8.1 Applications

The LLM Council framework for research assistance has a very practical applicability across various academic and professional domains :

1. **Academic Literature Review :** Researchers can pose a research topic to the council and get a synthesized overview that is based on various sources of knowledge and therefore the chances of omitting any of the important findings that a given model would fail to tell them are eliminated.

2. **Research Paper QA :** Abstracts or content can be pasted and domain-specific questions can be asked and a peer-reviewed answer can be received as opposed to interpretation of a single model.

3. **Educational Support :** The system can be used as a tutor, which is capable of analyzing different perspectives, capable of providing simple explanations of complex topics, provided by different models.

4. **Peer Review Assistance :** The evaluation framework of meta-model, which is a score given by meta-model based on Correctness, Clarity, Completeness, and Academic Depth of response. This clearly shows similarity with the structure of formal academic peer review..

8.2 Future Scope

1. **RAG Integration :** Connecting the system to existing live academic databases such as arXiv or PubMed via Retrieval-Augmented Generation would ground responses in real, cited papers rather than model training data alone.
2. **Dynamic Chairman Selection :** The meta-model could be chosen dynamically based on the query domain.
3. **Domain-Specific Councils :** Specialized councils could be created for medicine, law, or finance or any other specific domain, each of which contains models which are fine-tuned on domain data

9. CONCLUSION

This study presented the LLM Council, a multi-model deliberation framework applied to the domain of academic research assistance. By routing user queries through four language models, anonymizing their responses, and using a meta-model as an impartial judge, the system is able to mitigate few of the core imitations of single-model pipelines such as, hallucination, model-specific bias, and absence of self-evaluation. The architecture is based on two proven principles viz, ensemble learning and academic peer review, and applies them in a lightweight system. The formal assessment system, which gives a score to every answer in terms of Correctness, Clarity, Completeness, and Academic Depth, offers an interpretable and reproducible level of measurement of response quality.

The current implementation is a prototype, it demonstrates that multi-model deliberation is a good way forward to constructing systems that are more dependable, robust, and reliable AI-based research systems.

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